ACNS: Accessible Campus Navigation System

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Problem Statement

Campus Innovation – A real-time, crowdsourced map that actually works for wheelchair users.

Many campus maps do not consider accessibility needs such as ramps, elevators, and blocked pathways.
People with mobility challenges often face difficulties finding safe and accessible routes across campus.

The goal of this project is to create a system that helps users find accessible routes and also allows the campus community to report accessibility issues. By collecting real-time information from users, the system improves navigation and helps campus authorities quickly address accessibility problems.

System Architecture / Design

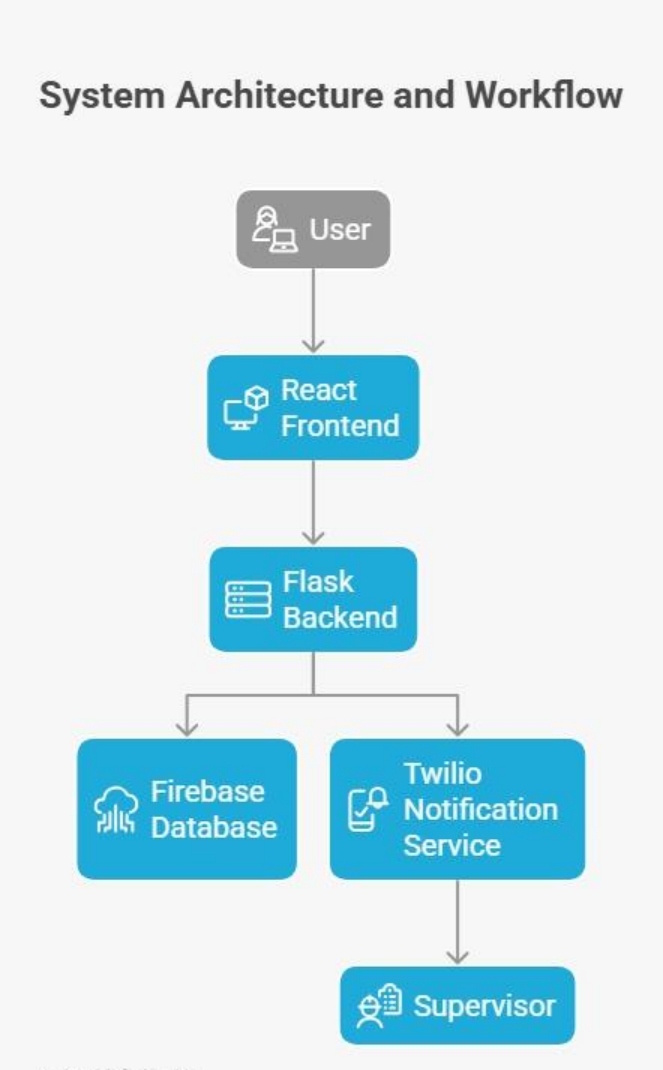
The system consists of two main parts: Frontend and Backend.

1. **Frontend (React Application)**
The frontend provides the user interface. It displays the campus map, allows users to select start and destination locations, and provides forms for reporting accessibility issues.

2. **Backend (Flask API)**
The backend processes requests from the frontend. It calculates accessible routes, manages issue reports, and interacts with the database.

3. **Database (Firebase Realtime Database)**
Firebase stores infrastructure nodes, issue reports, and gamification data.

4. **Notification Service (Twilio)**
When a new issue is reported, the system can send a WhatsApp notification to the supervisor.



Technologies Used

Frontend:

• React.js
• React Router
• Axios
• Mapbox GL
• Bootstrap
• Chart.js

Backend:

• Python
• Flask
• Flask-CORS
• Firebase Admin SDK

Database:
• Firebase Realtime Database

Other Tools:
• Twilio WhatsApp API
• Python Dotenv

Key Features and Functionality

• **Accessible Route Navigation**
Users can select a starting point and destination, and the system calculates the best accessible route.

• **Interactive Campus Map**
The campus map is displayed using Mapbox, showing routes and important locations.

• **Accessibility Issue Reporting**
Users can report problems such as broken elevators, blocked ramps, or damaged pathways.

• **Leaderboard System**
Users earn points for reporting issues and helping improve campus accessibility.

• **Admin Dashboard**
Administrators can view issue reports and analyze accessibility data.

• **Real-Time Notifications**
Supervisors can receive notifications when new accessibility issues are reported.

Explanation of How the Solution Works

1. A user opens the web application and views the campus map.

2. The user selects a start location and a destination.

3. The frontend sends a request to the backend API.

4. The backend uses the A* pathfinding algorithm to calculate the most accessible route based on available infrastructure nodes such as ramps, elevators, and stairs.

5. The backend sends the route data back to the frontend.

6. The route is displayed on the map using Mapbox.

7. If a user notices an accessibility issue, they can submit a report through the issue form.

8. The backend validates the report and stores it in Firebase.

9. The system can notify the supervisor through WhatsApp so the issue can be addressed quickly.

This process helps create a real-time, community-driven accessibility map for the campus.

